

$$1) y'' + \sin x = 0$$

$$\frac{dy}{dx} = p$$

$$\frac{dp}{dx} + \sin x = 0$$

$$\frac{dp}{dx} = -\sin x$$

$$dp = -\sin x dx$$

$$p = \cos x + C_1$$

$$dy = (\cos x + C_1) dx$$

$$y = \sin x + C_2 + x C_1$$

$$2) (1+x) \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = p$$

$$(1+x) \frac{dp}{dx} = -p$$

$$\frac{dp}{p} = \frac{dx}{-(1+x)}$$

$$\ln|p| = -\ln|1+x| + C_1$$

$$p = e^{-\ln|1+x| + C_1}$$

$$p = \frac{C_2}{\ln|1+x|}$$

$$\frac{dy}{dx} = \frac{C_2}{\ln|1+x|}$$

$$dy = \frac{C_2 dx}{1+x}$$

$$y = C_2 \ln|1+x| + C_3$$

$$3) \frac{d^2y}{dx^2} = e^{2x} + \cos 2x$$

$$p = \frac{dy}{dx}$$

$$\frac{dp}{dx} = e^{2x} + \cos 2x$$

$$dp = e^{2x} dx + \cos 2x dx$$

$$p = \frac{1}{2} e^{2x} + \frac{1}{2} \sin 2x + C$$

$$\frac{dy}{dx} = \frac{1}{2} e^{2x} + \frac{1}{2} \sin 2x + C$$

$$dy = \left(\frac{1}{2} e^{2x} + \frac{1}{2} \sin 2x + C \right) dx$$

$$y = \frac{1}{4} e^{2x} - \frac{1}{4} \cos 2x + Cx + C_1$$

$$4) \frac{d^2y}{dx^2} - 4 = + \left(\frac{dy}{dx} \right)^2 \quad \frac{dy}{dx} = p$$

$$\frac{dp}{dx} - 4 = p^2$$

$$\frac{dp}{dx} = p^2 + 4$$

$$dp = (p^2 + 4) dx$$

$$\frac{dp}{p^2 + 4} = dx$$

$$\frac{1}{2} \tan^{-1} \frac{p}{2} = x + C_1$$

$$\tan^{-1} \frac{p}{2} = 2x + 2C_1$$

$$\frac{p}{2} = \tan(2x + C_2)$$

$$p = 2 \tan(2x + C_2)$$

$$\frac{dy}{dx} = 2 \tan(2x + C_2)$$

$$dy = 2 \tan(2x + C_2) dx$$

$$y = -\ln|2x + C_2| + C_3$$

$$y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = \frac{dy}{dx} \quad p = \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{dp}{dx} = \frac{dp}{dy} \frac{dy}{dx} = \frac{dp}{dy} p$$

$$y \left[\frac{dp}{dy} p \right] + p^2 = p$$

$$yp \frac{dp}{dy} + p^2 = p$$

$$yp \frac{dp}{dy} = p - p^2$$

$$\frac{dp}{dy} = \frac{p - p^2}{yp} \rightarrow \frac{dp}{dy} = \frac{1-p}{y} \rightarrow \frac{dp}{1-p} = \frac{dy}{y}$$

$$\frac{dy}{y} + \frac{dp}{p-1} = 0$$

$$\ln y + \ln |p-1| = C_1$$

$$y(p-1) = C_1$$

$$p-1 = \frac{C_1}{y}$$

$$p = \frac{C_1}{y} + 1$$

$$\frac{dy}{dx} = \frac{C_1}{y} + 1$$

$$dy = \left(\frac{C_1}{y} + 1 \right) dx$$

$$\frac{dy}{\left(\frac{C_1+y}{y} \right)} = dx$$

$$\frac{y dy}{C_1+y} = dx$$

$$\int \left(1 - \frac{C_1}{C_1+y} \right) dy - \int dx = C_2$$

$$y - C_1 \ln(C_1+y) - x = C_2$$

$$y = C_1 \ln(C_1+y) + x + C_2$$